

Design System & UX

polycarpic — personal budgeting & wealth management (virtual Single Family Office)

Owner: ux-designer agent **Status:** Draft — Plan phase, matured alongside ARCH **Version:** 0.10

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1. Overview & Design Principles

What this product should feel like to use, and the principles that resolve design disputes.

- **Mobile-first when in doubt.** The primary v1 target is a self-hosted web dashboard used on desktop for deep review (statements, lot tables) and on mobile for quick checks (dashboard glance, confirming a draft transaction) — design the dense screens up from a mobile layout, not down from desktop.
- **Accessible by default (WCAG AA).** Non-negotiable per PRD §6. Financial gain/loss and status signals never rely on color alone — see [Financial semantics](#).
- **No decorative loading states.** Skeletons match the real layout; spinners are a last resort (see [Component inventory](#)).
- **Every number earns trust.** This is a general ledger, not a budgeting toy — a balance, a delta, or a pending-liability figure must always be traceable to its source (the journal entry, the lot, the reconciliation). The UI never shows a number it can't explain if clicked.
- **Immutability is visible, not just enforced.** A posted transaction reads as posted (badge, disabled edit affordance); a correction reads as a reversal + new entry, never as a silent edit (PRD §5).
- **Diverge from polycarpic's own tooling, deliberately.** The cairn tracker (our internal tool) and the product are not the same thing and shouldn't look like it — this system's identity is derived from **measured** screenshots and live style reads off Monarch Money and Origin (``useorigin.com``), not memory (Principal direction, 2026-09-28) — see [Inspiration: Monarch & Origin](#).
- **One brand hue, used honestly.** The Principal's neon-green brand color (`#2CFF05` , 2026-09-30 — "the project name relates to fruiting plants") measures 1.36:1 against white, far below AA — kept as the literal `--brand` value everywhere it clears AA (dark surfaces), with a darkened same-hue green for light-mode `--primary` , stated plainly rather than silently swapped for a different hue — see [Foundations](#).

2. Design System

The single source of truth for the visual language. The written spec lives alongside this doc; the machine-readable system lives in the two CSS files that `frontend-lead` consumes directly.

- `design-system-spec.md` § Inspiration — what was taken from Monarch Money and Origin, and why
- `design-system-spec.md` § shadcn-svelte reuse map (*pass 8, new*) — every component/region mapped to its shadcn-svelte component, block, or a stated "custom" reason
- `design-system-spec.md` — the written spec (token rationale, financial semantics, component inventory, accessibility)
- `tokens.css` — design tokens as CSS custom properties (hand-authored palette, not a vendored shadcn preset)
- `tokens.html` (*pass 6, new*) — every token decoded: name, value in both modes, swatch, one-line "used for"
- `screen.css` — component & screen styles built on the tokens

Figma stays the high-fidelity design tool for anything shipping pixel-exact (via the `ux-designer`'s Figma skills); `docs/DESIGN/` is the in-repo, code-level home that ships with the project and is the contract for implementation.

3. Design Tokens

Color, typography, spacing, radius, and shadow primitives — SvelteKit + shadcn-svelte (PRD §8). Edit values in [tokens.css](#); the swatches below preview the live variables. Full rationale in [design-system-spec.md](#) — or see [tokens.html](#) for every token decoded with both-mode values and a "used for" line.

3.1 Surfaces & Text

--background

--foreground

--card

--popover

--muted

--muted-foreground

3.2 Actions & States

--primary

--brand (dark surfaces only — see Foundations)

--brand-2 (pass 5, same rule — see Secondary accent)

--secondary

--accent

--destructive

3.3 Financial semantics (new — polycarpic-specific)

--positive

--warning

--destructive (reused for negative/danger)

See [design-system-spec.md § Financial semantics](#) — gain/loss, safe-harbor status tiers, and the "never color alone" rule.

3.4 Borders & Focus

--border

--input

--ring

3.5 Charts (sequential ramp — magnitude, not category)

--chart-1

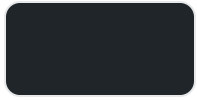
--chart-2

--chart-3

--chart-4

--chart-5

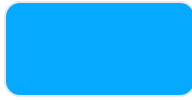
3.6 Sidebar (inverted, pass 4; re-hued blue, pass 5)



--sidebar



--sidebar-accent



--sidebar-ring

Dark-on-light in light mode, light-on-dark in dark mode — see [design-system-spec.md § Sidebar](#) for the collapse-to-rail behavior and three proposed expand/collapse cues.

Type: `--font-sans` *Inter* — one grotesque sans for body and headings, hierarchy by weight, confirmed against both reference products' actual app UI (their serif accent is marketing-only, see *Inspiration*); `--font-mono` *Roboto Mono* — Origin's own measured choice, for uppercase eyebrow labels, account numbers, tickers, tabular figures. **Radius:** measured `--radius-sm|md|lg|xl` = 6/8/12/20px (cards 12px, buttons 8px — both products' actual app-level values, not either's marketing-only pill).

4. Components & Screen Styles

The component inventory and the styles that realize it, built on the tokens. Styles live in [screen.css](#); full states-per-component table lives in [design-system-spec.md](#) § Component inventory.

Component	shadcn-svelte primitive	polycarpic usage
Net-worth hero	Chart (LayerChart, Area)	Big figure + signed delta on the figure + single-line area chart — direct Monarch pattern
Grouped account list	Table / custom	Type-grouped rows with sparkline + freshness — direct Monarch pattern
Segmented bar + legend	Custom, tokens-driven	Assets/Liabilities, allocation actual-vs-target — direct Monarch pattern
Badge	Badge	Entity/account/transaction status, link-health — see Financial semantics
Transaction table	Table	Account register, exceptions report, lots
Accordion (<i>pass 7, new</i>)	Accordion	Transactions' three urgency-ordered cards (Exceptions/Pending/Posted); Pending row expansion
Near-full-screen modal (<i>pass 7, new</i>)	Dialog	Journal-entry modal (header + legs) — manual entry, Correct entry, Split, Group & Summarize
Entity switcher	Tabs / Sidebar	Switch between Person/Trust/Business/Household
Transaction detail	Sheet	Drill into a posted/draft entry, its journal lines, source doc
Empty state	Card + plain text, no skeleton	Dashboard panels not yet shipped (tax at 1.0, allocation at M4) — see styled-screens/dashboard.html

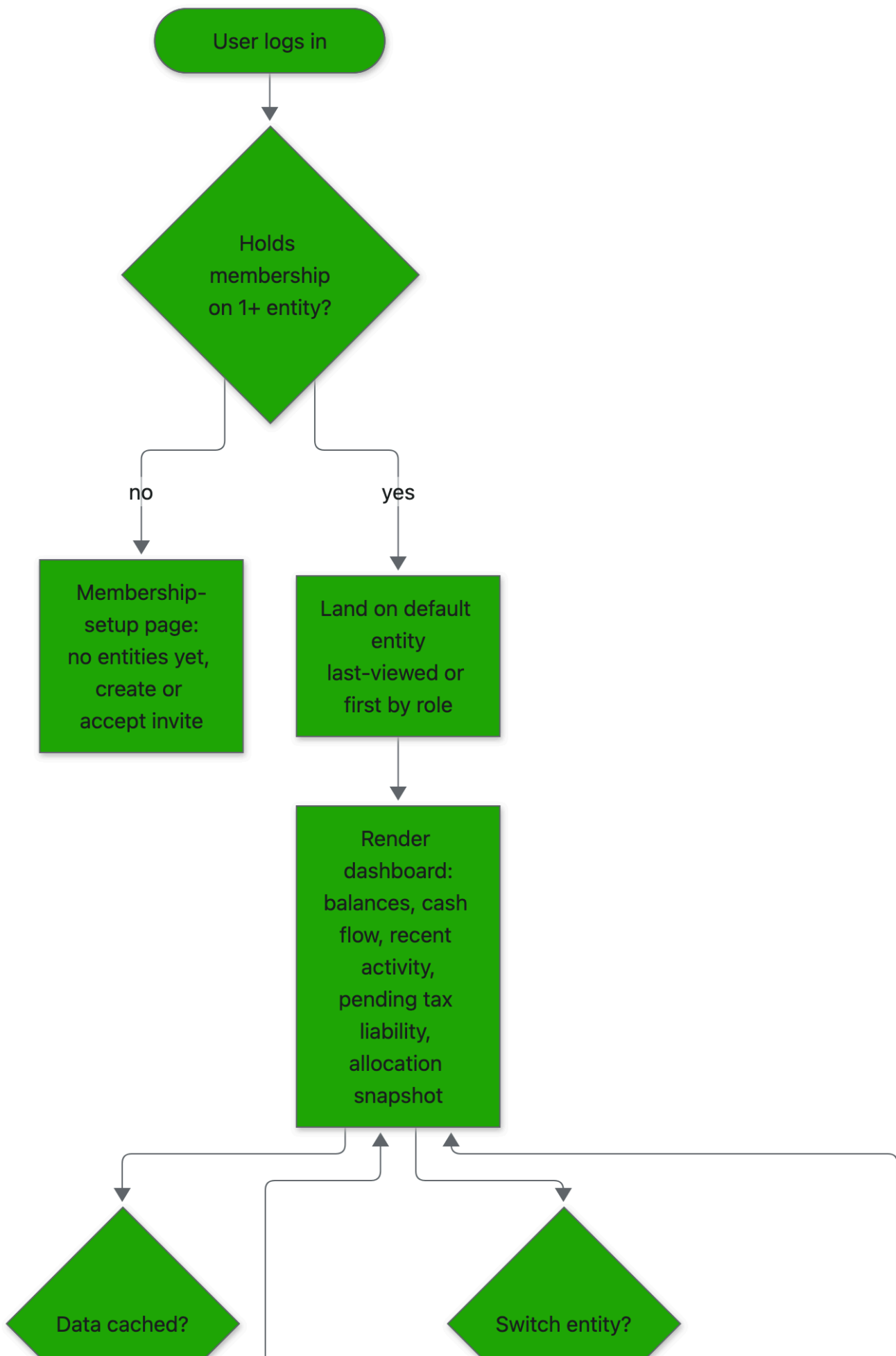
Component	shadcn-svelte primitive	polycarpic usage
Header (<i>pass 8, rebuilt</i>)	—, starts right of the sidebar	Expand/contract icon + breadcrumb left; entity selector, AI icon (greyed v1), search, notifications, settings gear right — see design-system-spec.md § Header
Sidebar (<i>pass 8, rebuilt</i>)	Sidebar (shadcn-svelte)	Viewport-fixed full height, seven nav items, brand top item, avatar-popover bottom item (Settings moved to the header gear) — see design-system-spec.md § Sidebar
Avatar cluster (<i>pass 5, new</i>)	—, custom	Overlapping initials in the entity selector, max 3 + overflow
Avatar popover (<i>pass 8, new</i>)	Popover / Dropdown Menu	User menu at the sidebar's bottom — profile, preferences, theme, keyboard shortcuts, Log out last
Public hero (<i>pass 4, corrected</i>)	—, custom	Home — pre-auth landing page, product statement + CTAs, not a signed-in screen
Split screen login (<i>pass 4, refined pass 5</i>)	—, custom	Login — full field list, centered form column, always-branded visual pane

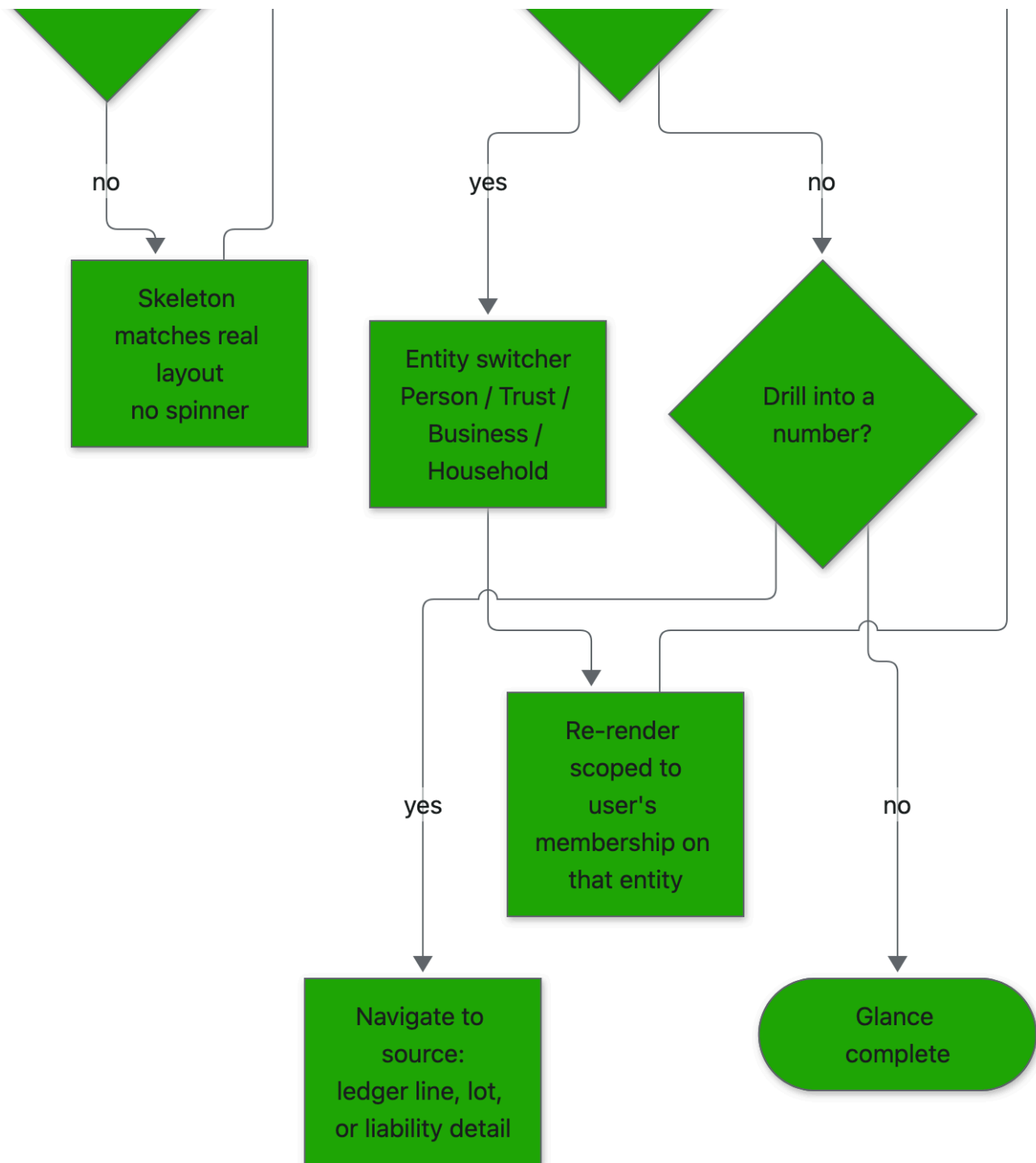
5. User Flows

How users move through the product to accomplish the highest-traffic stories (PRD §4). **Pass 5 fix (2026-09-30):** these previously linked to `flows/*.md`, which the browser serves as raw Markdown — the fenced `mermaid` blocks never ran through the page's own Mermaid loader, so nothing rendered (not a syntax error, as first suspected). Each flow is now embedded inline below; the `.md` files stay as the diffable source of truth in flows/.

5.1 Dashboard glance

Story (PRD §4.1): per-entity financial health at a glance. Source: flows/dashboard-glance-flow.md.



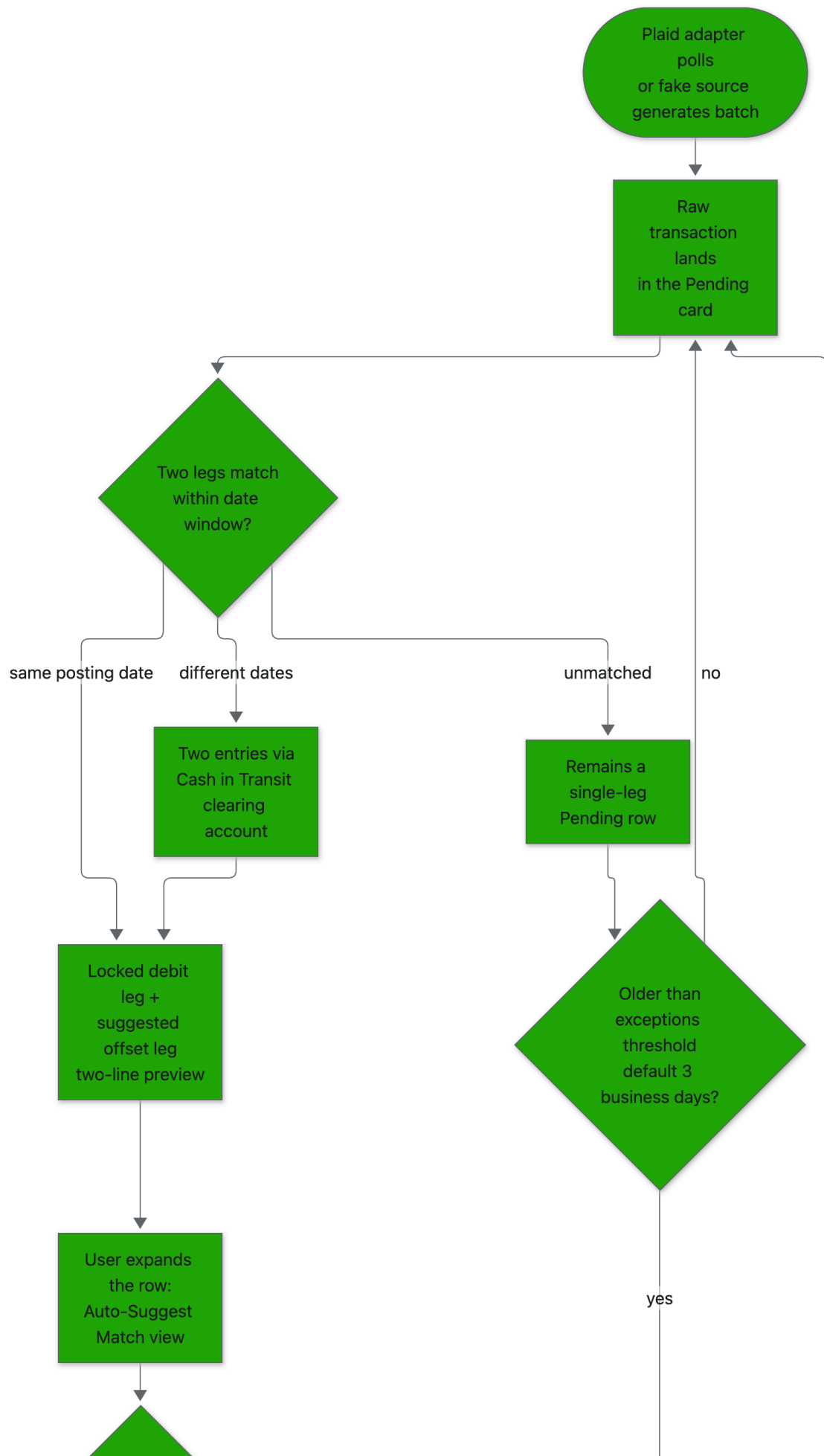


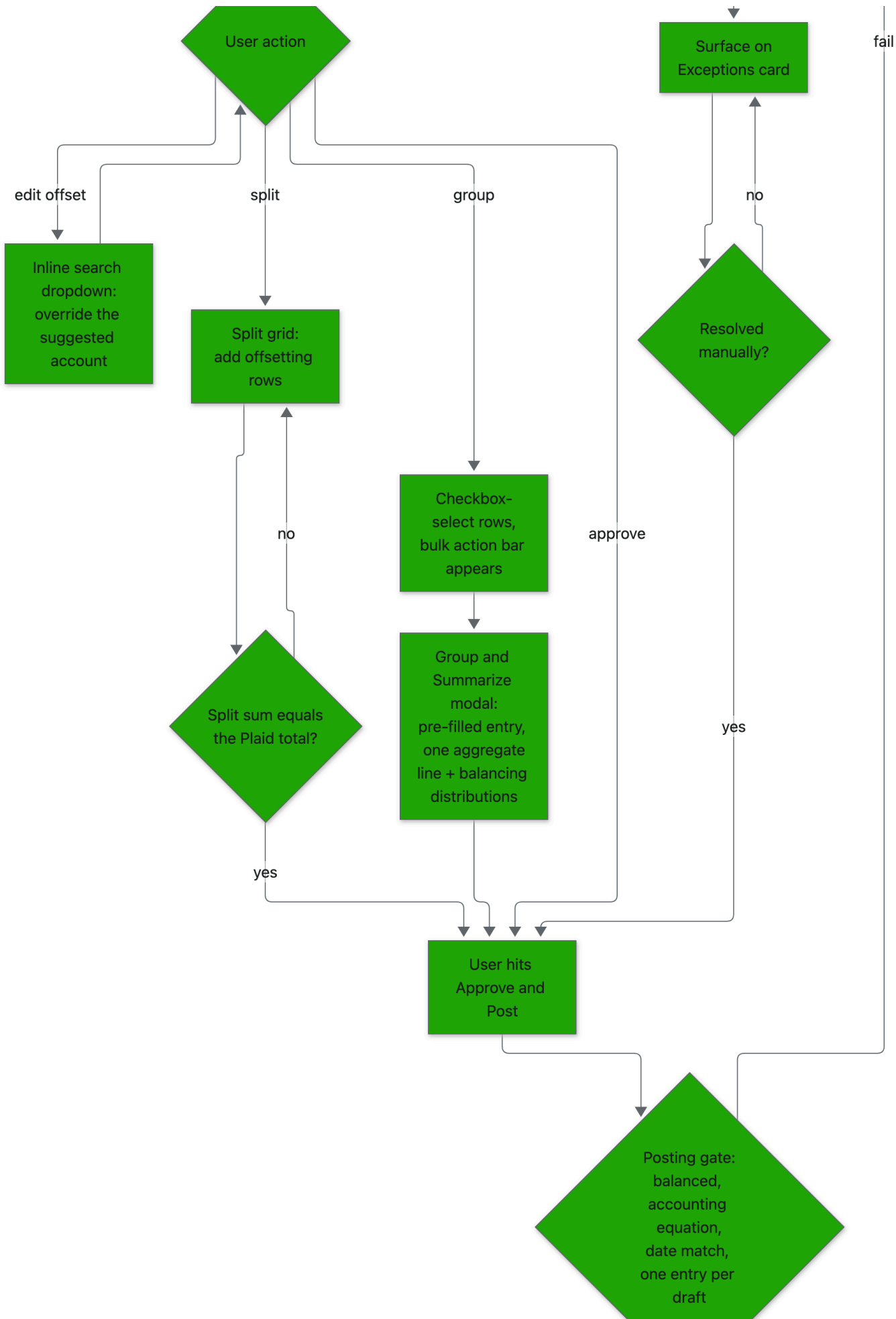
- Every number on this screen must be clickable through to its source (Overview principle: "every number earns trust").
- Filtered strictly by membership (owner/manager/viewer) — no transitive access via parent entity (PRD §5 permission model).
- **No-membership state (pass 6):** see [wireframes/membership-setup.html](#) — the Principal confirmed the dashboard's empty state can simply be the membership-setup page itself, not a separate illustrated empty-state screen.

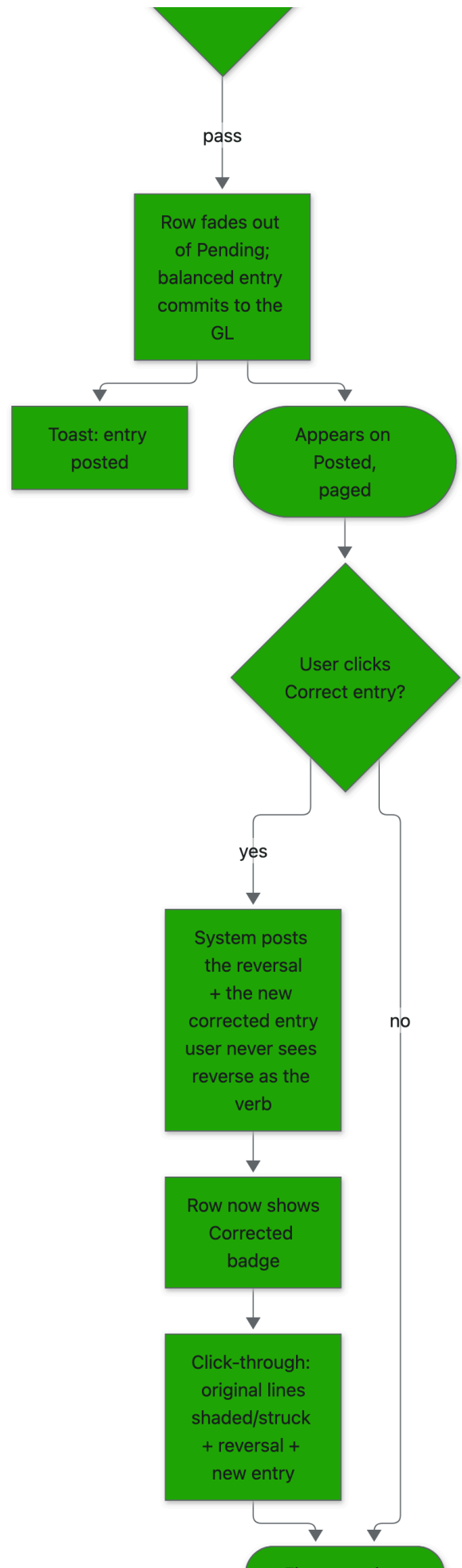
- **"Skeleton matches real layout, no spinner" (pass 6):** while dashboard data is loading, the screen shows placeholder blocks already shaped like the real cards (same card chrome, same approximate text-line widths) instead of a spinner — so nothing visually jumps or re-flows once the real numbers arrive, they just fade in where the placeholders already were.

5.2 Transactions — review, match & correct

Story (PRD §4.2): *staged for review → matched/split/grouped → posted → correctable. Source: [flows/import-confirm-flow.md](#). **Pass 7 full rewrite:** this screen is now **Transactions** ("Ledger"/"Import" renamed/retired) — the review mechanics below are the Principal's own Plaid bank-feed workflow spec.*







- **The locked leg** is always the custodial cash side (verified bank reality) — never editable, in the preview, Split grid, or Group modal. **The suggested offset leg** is pre-filled from historical rules + Plaid category, overridden in place via an inline search dropdown.
- **Split** is one draft → one multi-leg entry, live-validated to the Plaid total before it's submittable. **Group** is many drafts → one entry (the one many-to-one case, ARCH §3) via the shared journal-entry modal, pre-filled with an aggregate line + balancing distributions.
- **Posting gate:** $\Sigma \text{ debits} = \Sigma \text{ credits}$; accounting equation per entity; entry date matches bank date (or the CIT rule); every posted draft linked to exactly one entry — any failure keeps the item in Pending.
- **Reframed pass 6, unchanged:** a posted row's user-facing action is "Correct entry", never "edit"/"reverse" — reversal-only underneath (PRD §5). See design-system-spec.md § Correcting a posted entry.
- A corrected row carries the `Corrected` badge (`--outline` + `--brand-2` -tinted dot, pencil icon); click-through reveals the full double-entry trail — original lines shaded/struck, the reversal, and the new entry. An auto-posted row carries the same badge/action plus a `posted by rule - <name>` line (addendum, pass 6).
- Actor attribution (user / auto-post rule / system job) is shown on every draft and posted row, never inferred (PRD §5).

5.3 Tax liability & safe-harbor

Story (PRD §4.4): *pending liability, required payment, gap, pay.* Source: *flows/tax-liability-flow.md*.

Open Tax
Liability view
for entity +
jurisdiction

Running
accrual
annualizes
income to date
by tax treatment

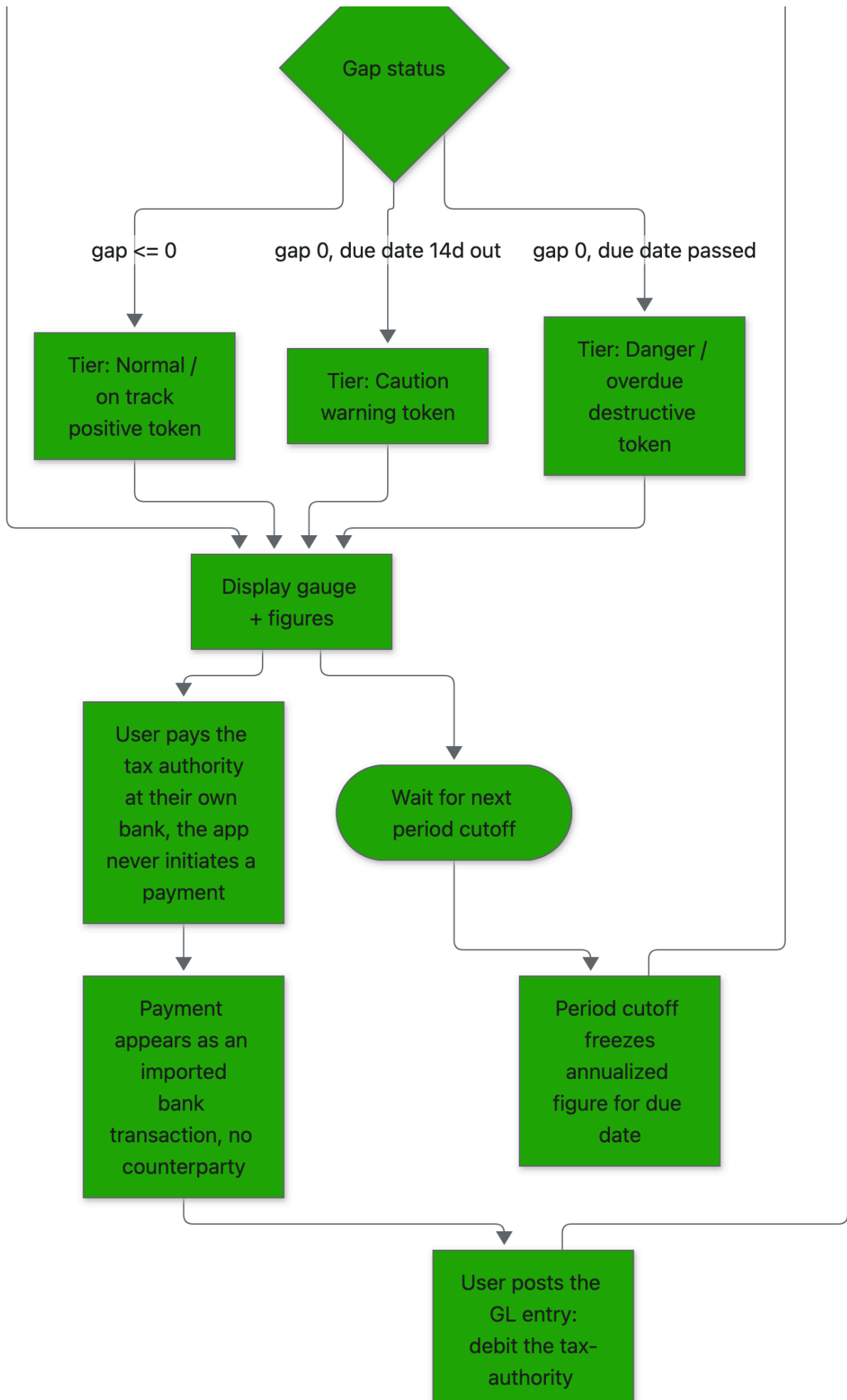
Two safe-
harbor bases
computed:
prior-year and
current-year

Required
payment to
date =
lesser of the
two safe-
harbor bases

Gap = required
payments to
date
minus custodial
balance,
are you current
with the
authority

Pending liability
=
annualized tax
to date minus
custodial
prepaid balance



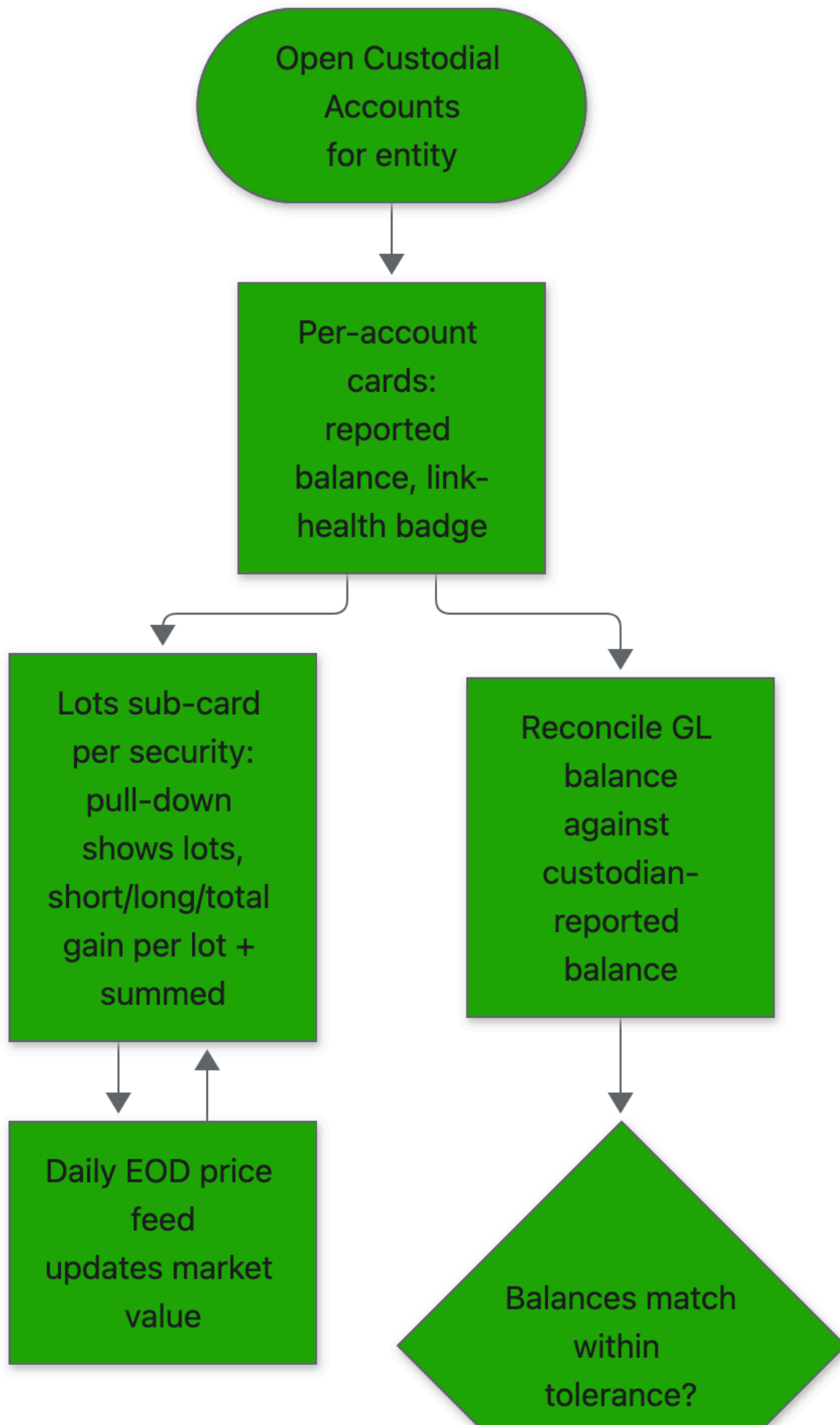


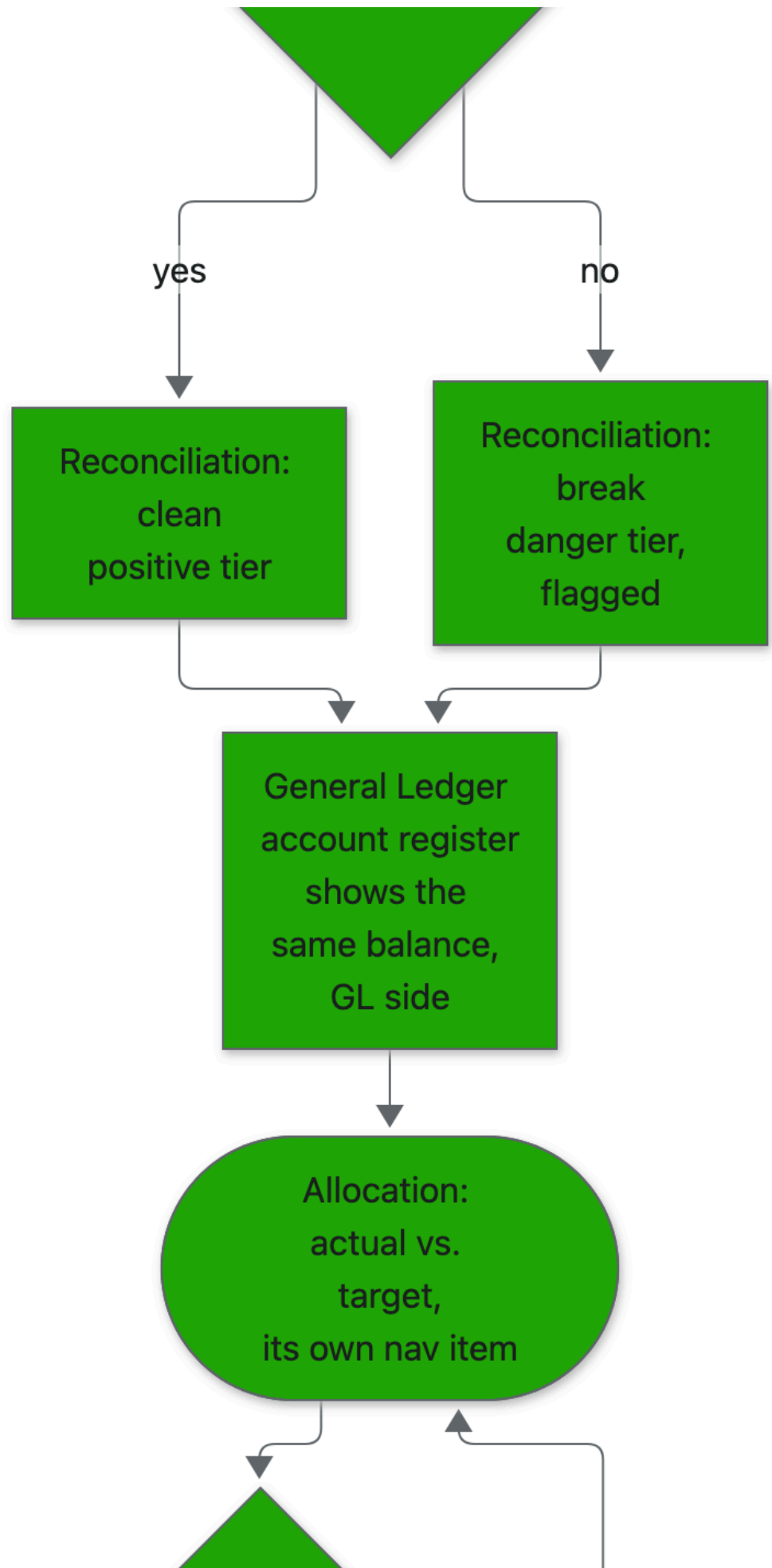
custodial
account,
pending liability
itself
does not
change, not
filed yet

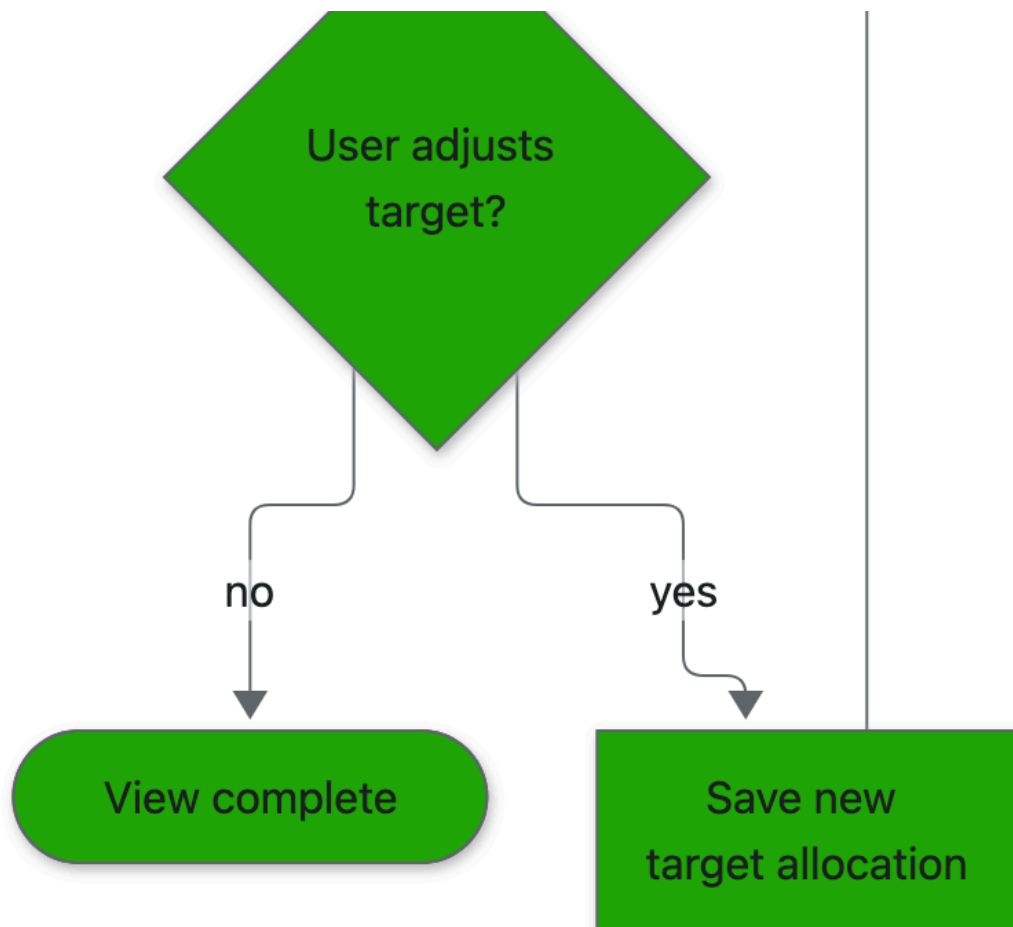
- The three-tier status (normal/caution/danger) reuses the product-wide status vocabulary — see [design-system-spec.md § Financial semantics](#).
- Jurisdiction due dates, multipliers, and cumulative percentages are data-driven, so this view must render correctly for an arbitrary number of jurisdictions (PRD §5).
- **Redrawn pass 6:** v1 never moves money — there is no "User initiates payment?" button. The user pays at their own bank; it lands as an imported transaction with no counterparty; the user posts the GL entry, debiting the custodial tax-authority account. Dashboard/tax-panel copy says "post your payment," never "pay" or "transfer."
- **Addendum, pass 6:** two figures, both measured against the custodial account's balance but not against each other — **Pending liability = annualized tax to date – custodial prepaid balance** (what you'd owe net of prepayments, if filed today); **Gap = required payments to date (safe-harbor) – custodial balance** ("are you current with the authority"). Gap is never required – pending .

5.4 Custodial reconciliation, lots & allocation

Stories (PRD §4.5–4.7): GL vs. custodial reconciliation, allocation vs. target. Source: [flows/securities-lots-flow.md](#). **Pass 7:** "Securities" is gone as a nav item — lots are now a Custodial Accounts sub-card, GL vs. custodial is two nav items not a toggle, Allocation is its own page.



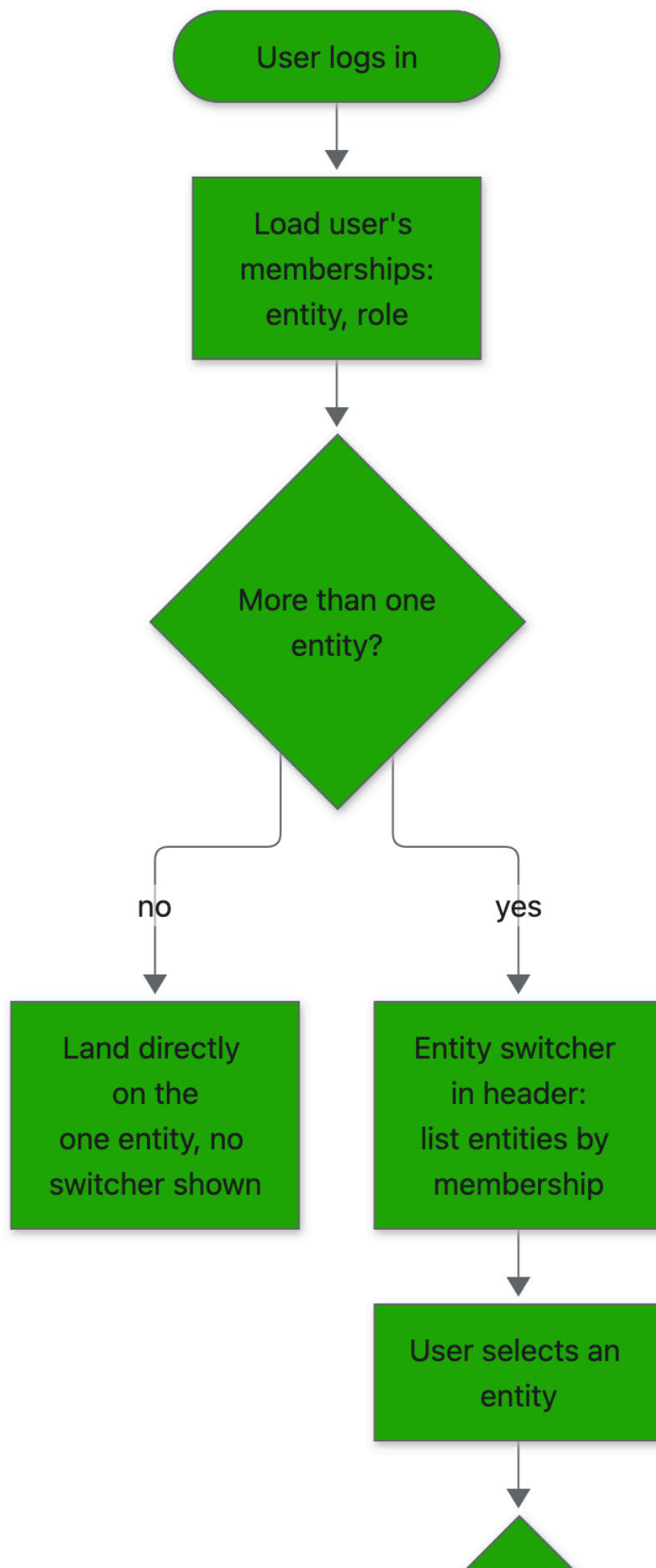


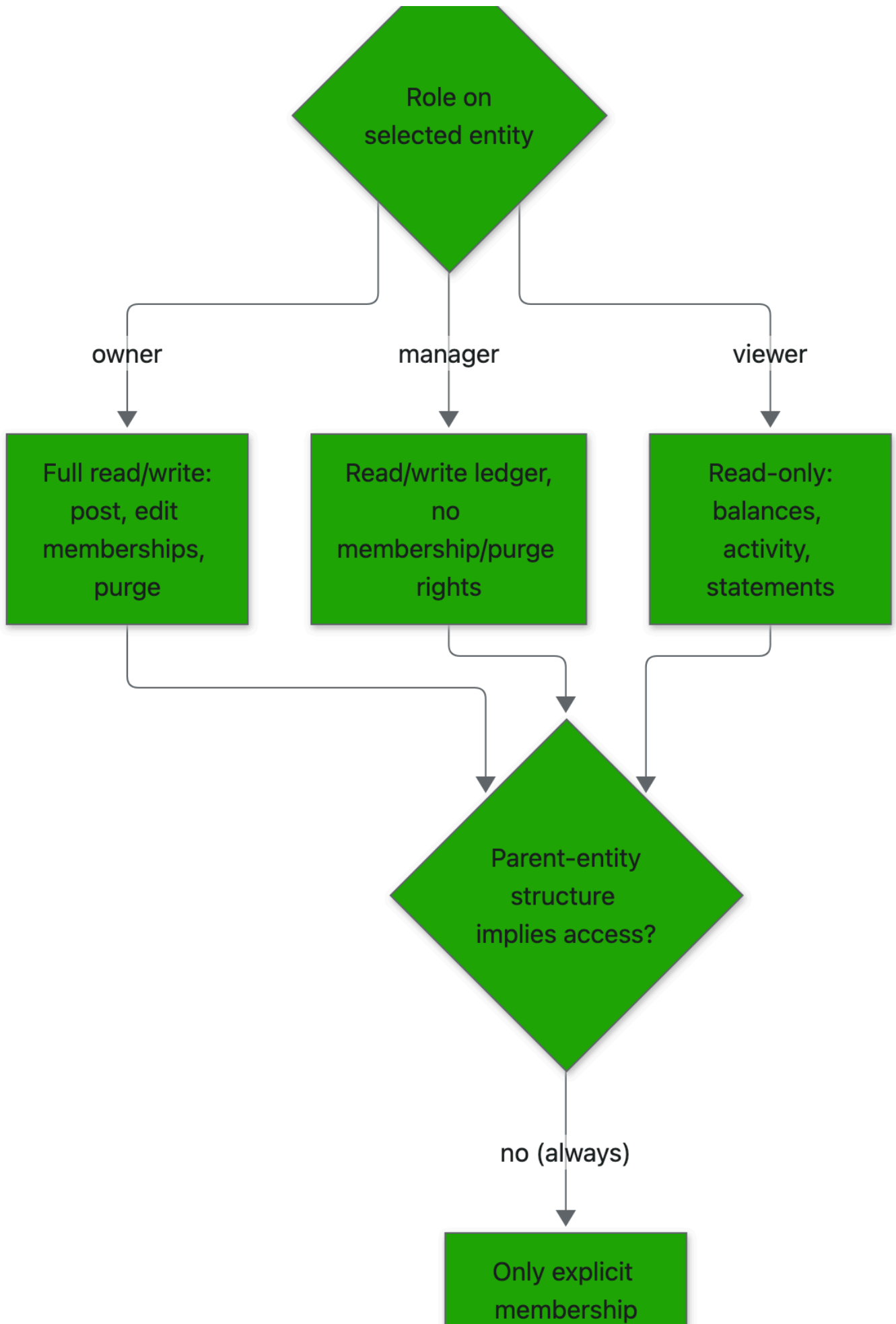


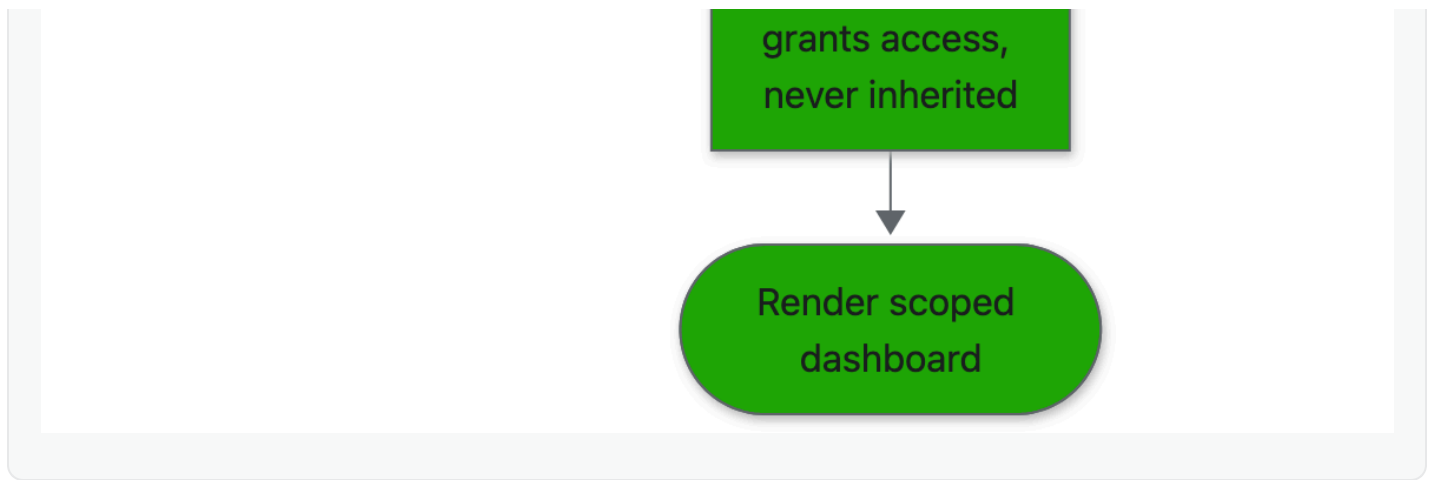
- **No toggle (pass 7):** General Ledger and Custodial Accounts are two separate nav items, not one screen behind a `Tabs` switch.
- Book-vs-market is implicit on the lots sub-card's own gain/loss figures, not a separate chart.
- A reconciliation break uses the danger tier (`--destructive`) and must surface on the Dashboard glance too.

5.5 Entity switch & membership access

Story (PRD §4.8): *owner/manager/viewer boundary. Source: [flows/entity-membership-flow.md](#).*







- A Household's membership never transitively grants access to a member Person entity, or vice versa (PRD §5, hard requirement).
- A viewer never sees a write affordance rendered-then-disabled — write controls are omitted entirely, not shown-and-greyed.

6. Wireframes

*Low-fidelity layout of each key screen — structure before style. Assets in [wireframes/](#). **Pass 7 (2026-10-02):** reorganized for the eight-item sidebar ([design-system-spec.md § Sidebar information architecture](#)) — rolling out across stages 2–5 of the same pass; items not yet rebuilt are marked pending rather than linked to a stale file.*

- [Dashboard](#) — per-entity financial health
- [Transactions](#) (*stage 2, done*) — three urgency-ordered accordion cards (Exceptions/Pending/Posted), Pending inbox built around the Plaid review/match workflow.
- [General Ledger](#) (*stage 3, done*) — chart-of-accounts tree; each account opens its own [register](#) page.
- [Custodial Accounts](#) (*stage 4, done*) — per-account cards, a lots sub-card (pull-down per security, short/long/total gains), and the `[+ New]` provider-item modal (aggregator/manual/file, discovered-account mapping, token refresh, unlink).
- [Allocation](#) (*stage 5, done*) — actual-vs-target, promoted out of the old Securities screen.
- [Reports](#) (*stage 5, done*) — Balance Sheet / Income Statement / Cash Flow, low fidelity; empty state until POLY-M5.
- [Entities](#) — unaffected by the sidebar rework
- [Settings](#) (*stage 5, done*) — batch-import schedule/options + app preferences.
- [Home](#) (*pass 4, corrected*) — public pre-auth landing page
- [Login](#) (*pass 4, new; corrected pass 6*) — split screen, centered form pane
- [Membership setup](#) (*pass 6, new*) — dashboard first-run / no-membership empty state, linked from flow 5.1
- [Sidebar expand/collapse cues](#) (*pass 4, decided pass 5*) — option A (chevron) chosen, B/C kept for the record; unaffected by the pass-7 item-list change

6.1 Operator surfaces

Not reachable from any in-app nav, not part of the eight-screen IA — operator-only, reached by SSH tunnel to the host.

- [Admin](#) (*pass 7 addendum, 2026-10-02*) — **placeholder, pending PRD/ARCH/SECURITY definition (POLY-81)**. A plain status page (services, versions, DB/backup status, last sync, telemetry receiver state) + an "Administrative actions (TBD)" link list.

Gone as of pass 7 (stage 4): [wireframes/securities.html](#) and [wireframes/import.html](#) are retired — their content is absorbed into Custodial Accounts (lots sub-card, provider-item modal) above.

7. Styled Screens

High-fidelity screens that apply the design system to the wireframes, consuming `tokens.css` + `screen.css` — a visual regression reference and a head-start for `frontend-lead`. Prioritized by traffic (PRD §4 stories 1 & 2); remaining screens promote from wireframe fidelity as the design system stabilizes. **Pass 7:** only Dashboard and Transactions get styled-screen fidelity for now, same traffic-priority rule as before — General Ledger, Custodial Accounts, Allocation, Reports, and Settings stay at wireframe fidelity until there's evidence a styled build is needed before Implement (judgment call, flagged in Open Questions).

- Dashboard
- Transactions (stage 2, done)
- Home (pass 4, new)
- Login (pass 4, new)

8. Accessibility

Accessibility is a design responsibility, not a QA afterthought. Full table in [design-system-spec.md § Accessibility](#).

Concern	Standard / approach
Color contrast	WCAG AA (4.5:1 text, 3:1 large/UI) — every token pair re-measured for the pass-4 neon-green rebrand, both modes; <code>--brand</code> deliberately fails against light surfaces (1.36:1, by design — never used there, see Foundations); no CVD simulation run yet (open item, now higher-priority given how central the brand hue is)
Color-alone signaling	Never — every gain/loss/status pairs color with a sign/icon/label
Keyboard	All interactive elements reachable & operable (shadcn default)
Screen readers	Semantic markup; labelled controls; delta sign readable from text content
Motion	Respect <code>prefers-reduced-motion</code> (policy not yet written)

9. Open Questions

Unresolved design decisions and assumptions that could be wrong. Batched to team-lead for relay to the Principal — not blocking.

- **Resolved 2026-10-02 (pass 8):** app shell rebuilt on the shadcn Sidebar component's own pattern (Principal review 5) — sidebar now fills the viewport height (`position: fixed` , not `sticky` , fixing a real bug where a wireframe's own explanatory note pushed the sidebar below the visible viewport), brand moved to the sidebar's top item, an avatar-popover user menu (profile/preferences/theme/shortcuts/Log out) replaces the old bottom chevron, and Settings left the sidebar for a header gear icon (seven nav items remain). Header now starts right of the sidebar: expand/contract icon + breadcrumb left, entity selector/AI-icon/search/notifications/gear right — screens no longer repeat their title as an on-page heading. New § shadcn-svelte reuse map answers the Principal's reuse question directly. Applied to screen.css, wireframe.css, both styled screens, and all nine sidebar-bearing wireframes.
- **Resolved 2026-10-02 (pass 7, all 5 stages):** sidebar reworked to eight items per the Principal's interview (`temp/2026-10-02-sidebar-interview.md`) — Dashboard, Transactions, General Ledger, Custodial Accounts, Allocation, Reports, Entities, Settings. "Securities" and "Import" are gone. Transactions rebuilt around the Plaid review/match workflow (expansion, Split, Group, a shared journal-entry modal); General Ledger opens to a chart-of-accounts tree with a per-account register page; Custodial Accounts gained a lots sub-card and a provider-item modal; Allocation, Reports (empty until POLY-M5), and Settings are new. See design-system-spec.md § Sidebar information architecture onward. **Judgment call:** only Dashboard and Transactions got styled-screen fidelity, matching the existing traffic-priority rule — flag if General Ledger, Custodial Accounts, or Allocation need a styled build sooner.
- **Resolved 2026-10-01 (pass 6):** `--warning` re-hued to a true orange with a dark ink (was brown-reading); light-mode chart ramp brightened; tokens.html added; Login wireframe corrected to match the centered styled screen; dashboard first-run/empty-state wireframe added (`membership-setup.html`); skeleton-vs-spinner explained in plain words; flow 5.2 reframed around the user-facing "Correct entry" action; flow 5.3 redrawn to remove the app-initiates-payment step.
- **Resolved 2026-09-30 (pass 5):** sidebar expand/collapse cue — option A (chevron) chosen; sidebar now full-height; header restructured to brand lockup + entity selector with an avatar cluster; login form pane centered — design-system-spec.md § Header, § Sidebar, § Login.
- **Avatar cluster overflow (pass 5):** the "+N" fold for 4th+ members is specified but not yet shown live in either reference screen (both currently have ≤ 2 members) — worth a quick visual

check once a real multi-member entity exists.

- **Login's visual pane (pass 4):** built as a brand gradient, not a photo — no photography asset exists for this product. A real photo/illustration asset (in the style of the captured `dribbble` reference) would likely read stronger; flagging as a design-asset gap, not a layout question.
- No CVD (color-vision-deficiency) simulation run on the new palette yet — now higher priority given how central the brand hue is (recommended before lock, design-system-spec.md § Accessibility).
- **Superseded by pass 7:** the former Accounts screen's grouped-account-list + segmented-bar treatment was only built on the Dashboard styled screen — Accounts itself is now split into General Ledger and Custodial Accounts (both wireframe fidelity, stages 3–4); promote either the same way as a fast-follow if the gap matters before Implement.
- Native iOS/macOS token export format (kickoff §2.10) — not yet decided, non-blocking for web.
- `prototype/` throwaway UX prototype (kickoff §2.9) not yet built — confirm ownership/timing with team-lead.

10. Appendix

- PRD user stories & functional requirements: <docs/PRD/index.html> §4–5
- Kickoff record, frontend/design system decisions: docs/project_kickoff.md §2.9

10.1 FAQs

Q: How do I add review comments to this design doc?

A: Hand-edit `docs/DESIGN/comments.md` with `## §<section-id>` blocks, or run `/serve-docs DESIGN` and use the inline + Comment widget. Then run `/refine-doc DESIGN`. See <process/WORKFLOW.md> → Doc review loop.

Q: Why is the design system in `docs/DESIGN/` and not in the app's source?

A: It's the design *deliverable* — the contract. During Implement, `frontend-lead` imports or derives the app's real stylesheets from `tokens.css` / `screen.css`.

Q: Where did `variants.css` go?

A: It was generated infrastructure for the `cairn` tracker tool's own theme switcher, never part of the product — removed from this directory (POLY-78 relocates its generator target the same way). See <design-system-spec.md> § Provenance & shared tooling.

